AUTOMATIC DOOR CONTROL USING PLC

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ABSTRACT

Automatic door systems have become an essential part of modern infrastructure, providing convenience, safety, and energy efficiency in public and private spaces. This project presents the design and implementation of an automatic door open and close system using a Programmable Logic Controller (PLC). The system utilizes sensors such as infrared (IR) sensors to detect the presence of a person and sends signals to the PLC. Based on the programmed logic, the PLC controls the operation of a motor through relays, enabling the door to open and close automatically.

KEYWORD : PLC, Automatic Door System, Motion Sensor, Motor Control, Interlocking Logic, Ladder Programming.

INTRODUCTION

Automation plays a significant role in modern industrial and commercial systems by improving efficiency, safety, and reliability while reducing human effort. One of the most widely used technologies in automation is the Programmable Logic Controller (PLC). A PLC is an industrial digital controller designed to operate machines and processes through programmed logic. It is preferred over traditional relay-based systems because it is flexible, easy to modify, simple to troubleshoot, and highly reliable.

The Automatic Door Open and Close Using PLC project focuses on developing a system that automatically operates a door based on human presence. Automatic doors are commonly used in shopping malls, hospitals, airports, offices, and hotels. They provide convenience, improve hygiene by reducing physical contact, and help in energy saving by ensuring the door remains closed when not in use.

In this system, sensors such as infrared (IR) or motion sensors are installed near the entrance to

detect a person approaching the door. When a person is detected, the sensor sends a signal to the PLC. The PLC processes this input according to the programmed instructions and activates a motor to open the door. After a preset time delay or when no presence is detected, the PLC automatically sends a command to close the door.

To ensure safe operation, interlocking logic is included in the PLC program. This prevents the motor from running in both forward and reverse directions simultaneously, protecting it from damage. Additional safety features such as obstacle detection, overload protection, and an emergency stop button are incorporated to enhance system reliability and user safety.

The system is thoroughly tested and debugged to verify sensor response, timing accuracy, motor

performance, and smooth door movement. Overall, this project demonstrates the practical application

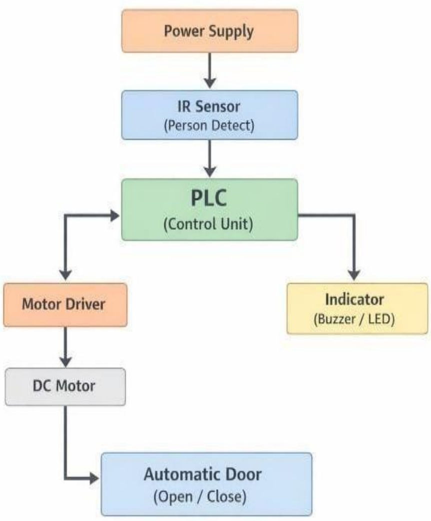
International Scientific Journal of Engineering and Management (ISJEM) ISSN: 2583-6129
Volume: 05 Issue: 04 |Special Edition | April – 2026 DOI: 10.55041/ISJEM06063
An International Scholarly || Multidisciplinary || Open Access || Indexing in all major Database & Metadata

of PLC in automation and provides a simple, efficient, and reliable solution for automatic door control in real-world environments.

METHODOLOGY

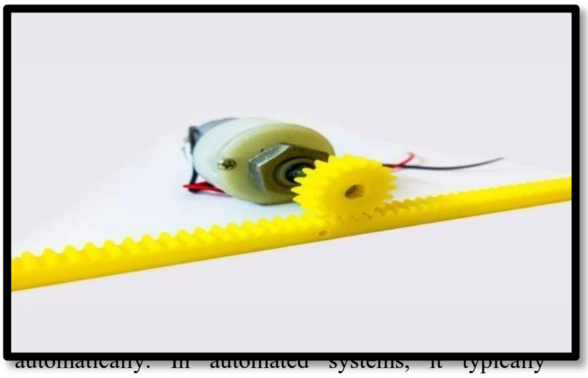
The development of a PLC-based automatic door open and close system begins with designing the system architecture by dividing it into three main sections: input, control, and output. The input section includes sensors such as IR or proximity sensors to detect human presence, while the control section consists of the PLC that processes input signals and executes the programmed logic. The output section includes the motor, relays, limit switches, and the mechanical door mechanism. A block diagram is prepared to show the signal flow from sensors to the PLC and then to the motor. During hardware implementation, sensors are installed at suitable positions, the PLC is mounted in a control panel with proper wiring, and the motor is mechanically connected to the door using gears or sliding mechanisms. Limit switches are fixed at fully open and closed positions to prevent over- travel. The PLC is programmed using ladder logic to open the door when motion is detected, hold it open for a preset time, and then close it automatically.

BLOCK DIAGRAM



COMPONENTS USED

he Siemens S7-1200 is a compact and versatile programmable logic controller designed for small to medium industrial automation applications such as automatic doors, conveyors, and machine control. It features an integrated CPU with built-in digital and analog I/O, supports PROFINET communication, and can be expanded with additional modules. Programmed using TIA Portal in languages like Ladder Diagram and Function Block Diagram, it processes input signals from sensors and controls output devices efficiently. Due to its compact design, flexibility, and ease of use, it is widely used in modern automation systems, though it is less suitable for very large- scale processes.



automatically. In automated systems, it typically includes a motor, sensors (such as IR sensors), limit switches, and a controller like a PLC. When a sensor detects a person or object, it sends a signal to the controller, which activates the motor to open the door. Limit switches ensure the door stops at fully open or closed positions, and after a

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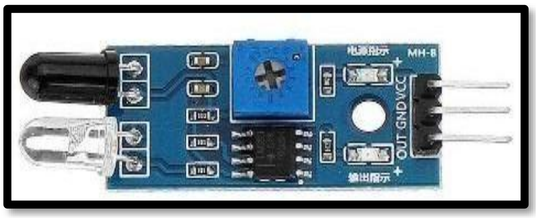
set time or when no presence is detected, the controller closes the door. Such mechanisms are widely used in malls, hospitals, and offices for convenience, improved hygiene, and enhanced safety.



An Infrared (IR) sensor is an electronic device used to detect the presence or movement of objects by emitting and receiving infrared radiation. It typically consists of an IR LED (transmitter) and a photodiode or phototransistor (receiver). The transmitter emits infrared light, and when an object comes in front of the sensor, the light reflects back and is detected by the receiver, generating a signal. IR sensors are widely used in automation and control systems for object detection, obstacle sensing, and proximity measurement. They are commonly used in applications such as automatic door systems, line-following robots, security systems, and counting systems. IR sensors are cost-effective, easy to interface with controllers like PLCs and microcontrollers, and provide quick response

and surface color of objects.

CIRCUIT DIAGRAM

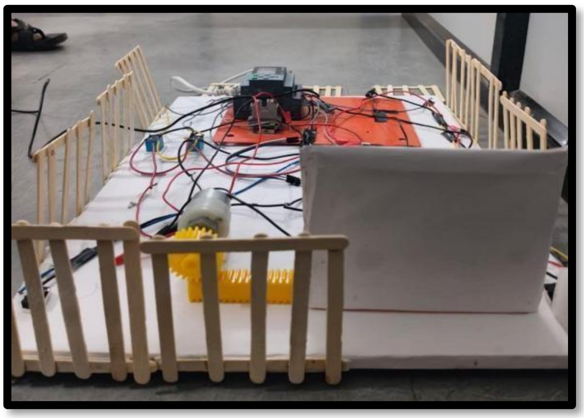


Fig.1. Door is Closed

RESULT

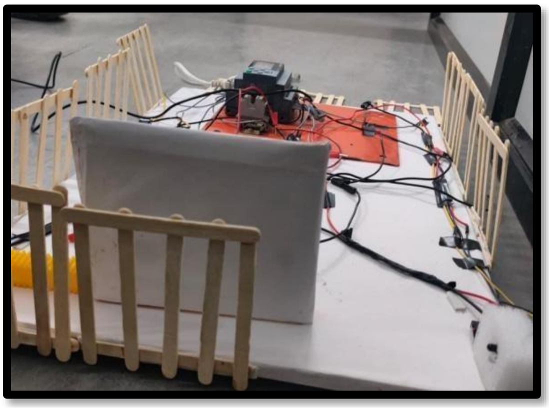


Fig.2. Door is Open

The experimental results of the PLC-based automatic door system demonstrate its performance and reliability under real-time operation. When a person approaches the door, the infrared or proximity sensor detects their presence and sends a signal to the PLC. Upon receiving this signal, the PLC energizes the motor to open the door. The door opens smoothly and stops precisely at the fully open the PLC starts a timer, keeping the door open for a preset duration to allow safe passage. Once the timer completes, the PLC reverses the motor to close the door. The door closes smoothly and stops at the fully closed position when the closed limit switch is activated. During multiple cycles of operation, the system consistently performed as expected. The interlock logic ensured that the motor never ran in both directions simultaneously, preventing mechanical damage. Overall, the system demonstrated precise control, smooth operation, and reliable automatic opening and closing of the door. The experimental results confirm that the PLC program and hardware setup work efficiently for real-time automatic door control.

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ADVANTAGES

1. Hands-free and convenient operation
2. Improved hygiene (no physical contact)
3. Enhanced safety with interlocking and sensors
4. Energy efficient (reduces unnecessary door opening)
5. Reliable and accurate control
6. Easy to modify and troubleshoot
7. Reduced human effort
8. Suitable for high-traffic areas

DISADVANTAGES

1. Higher initial installation cost
2. Requires technical knowledge for programming and maintenance
3. Dependence on continuous power supply
4. Sensor malfunction may affect operation
5. Maintenance and repair costs can be high
6. System failure may cause inconvenience in busy areas

APPLICATION

1. Hospitals – For hygienic, touch-free entry
2. Shopping malls – Main entrance sliding doors
3. Banks – Secure access control
4. Industries – Ware house and loading bay doors
5. Metro stations & Airports – Platform doors
6. Hotels – Automatic lobby doors

CONCLUSION

The PLC-based automatic door system is an advanced automation solution that integrates PLCs, sensors, timers, motors, and limit switches to ensure smooth, safe, and efficient operation. It detects a person's presence, opens the door, keeps it open for a set time, and closes it securely, while interlock logic protects the motor from damage. Testing shows reliable performance, fast sensor response, and consistent operation. With advantages such as hands-free access, improved hygiene, safety, and energy efficiency, the system is suitable for hospitals, airports, offices, malls, industries, and smart homes. Overall, it enhances convenience and reliability, and future technological integration will make it even more efficient and intelligent.

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